

Responses to the feedback on the thesis

1. I have serious concerns regarding the overall quality, clarity, and flow of the thesis. The writing style and organization make the document difficult to read and, at times, frustrating to follow. Important details are not always easy to locate. For instance, Annexure 8.13—referred to multiple times across different chapters—could not be found in the submitted document. Similarly, Appendix 7.4.2 appears to be missing. Statements such as “The detailed process of designing the card deck is discussed in Section 3.3 of the publication are unclear and confusing, particularly when the referenced material is not adequately presented within the thesis itself. There are several similar instances throughout the document.

Additionally, the sequencing of content is often problematic. In some sections, interpretations are presented before the results are described, which disrupts the logical flow of the analysis. For example, on page 37 in Chapter 4, interpretations appear before Table 4.1, whereas the methods and results are discussed only after the table. Such inconsistencies in structure make it difficult to follow the argument and assess the findings systematically.

Response:

The sections are arranged. All links to sections, chapters, and appendices are verified and checked for consistency.

2. My primary concern relates to the methodology of the study. The thesis does not clearly explain how the participants were selected. It is also unclear how they were allocated to the experimental and control groups. Was randomization used to assign participants to the intervention and control groups? If not, was any matching procedure adopted? Furthermore, was any attempt made to compare the baseline characteristics of participants across the two groups? Were the characteristics adequately balanced between the intervention and control groups? In the absence of such critical methodological details, it is difficult to assess the internal validity of the study. If the groups were not comparable at baseline, relying solely on a simple t-test would be insufficient to draw valid inferences. These methodological gaps significantly limit confidence in the reported findings. This concern is particularly relevant to Chapters 4, 5, and 6.

Response:

Participants' recruitment and grouping are added for all the studies.

We tried to balance each group in age, education, and experience.

Demographic details of participants and the detailed breakdown of grouping information are shown in Table 4.1 (Chapter 4), Table 5.1 (Chapter 5), and Table 6.1 (Chapter 6).

The pretest scores in all 3 studies show baseline equivalence scores for all groups compared.

3. The thesis also employs qualitative methods; however, no clear justification is provided for their inclusion. It is not explained why Focus Group Discussions (FGDs) were considered necessary within the overall research design. The process of selecting participants for the FGDs is also unclear. Furthermore, the thesis does not describe how the FGDs were analyzed, whether any systematic analytical framework was adopted, or whether any qualitative data analysis software was used. Equally important, it is not evident how the insights generated from the FGDs were integrated into the broader analysis. The thesis does not clearly explain how the qualitative findings informed the interpretation of results or contributed to the overall research objectives. In the absence of these methodological details, it is difficult to interpret the qualitative findings with confidence.

Response:

Sections 4.1 (2a) and 6.2 briefly describe the FGDs for Studies 1 and 3. Regarding the inclusion of Focus Group Discussions (FGDs), their primary purpose is to complement quantitative findings by capturing participants' experiences, perceptions, and reflections that cannot be fully understood through quantitative measures alone. Qualitative insights were intended to provide deeper contextual understanding and support the interpretation of observed outcomes. The Braun and Clarke method of thematic analysis was followed, and the analysis was conducted manually without software such as NVivo. FGDs also helped as an icebreaker with participants.

4. In Chapter 4, the scholar states that “The AR-based game was more interesting and effective in retaining knowledge significantly, even after three months.” However, it is unclear how knowledge retention after three months was assessed if the post-test was conducted immediately after the session. The thesis does not provide sufficient clarity regarding the timing and design of follow-up assessments to substantiate this claim. In the same chapter, it is also mentioned that “The AR-based game showed effective, immediate knowledge gains. However, the physical card game showed better knowledge retention.” These statements appear to contradict the earlier assertion regarding the AR-based game’s superiority in knowledge retention. The statistical analyses supporting these conclusions are not clearly presented, making it difficult to understand the basis for the interpretations discussed on page 38.

Response:

There was confusion about this study's reporting with other studies.

This study does not check for retention.

Parts of Chapter 4 have been corrected.

5. Furthermore, in ‘Study 1’, the inclusion of a placebo or standard refresher training control group would have been essential to draw a robust conclusion that the card game was more effective than conventional refresher training methods. In the absence of such a comparison group, the validity of the claim that the card-based intervention is superior to standard training remains questionable.

Response: I agree that the absence of a placebo comparison limits the strength of claims regarding the effectiveness of the card-based intervention relative to conventional methods. However, this limitation is acknowledged at the end of Chapter 4. Statements that implied superiority over standard refresher training were removed, and instead, our findings were framed as preliminary evidence of potential effectiveness.

6. In Chapter 5, the scholar concludes that the difference between the digital card game and the physical card game groups was statistically significant. However, the corresponding t-test results supporting this claim are not presented in Table 5.1. On the contrary, my reading of Table 5.1 suggests that there is no statistically meaningful difference between the digital and physical card game groups. This discrepancy raises concerns regarding the accuracy of the statistical analysis and the validity of the interpretations drawn from it. Furthermore, I was unable to locate any statistical analysis conducted to test Hypothesis H2. The absence of clearly reported results for a stated hypothesis is a serious omission and further undermines confidence in the analytical rigor of the thesis.

Response:

I respectfully clarify that the examiner’s interpretation of Table 5.1 does not fully reflect the study’s results. In response to this comment, I have revised Chapter 5 to improve clarity. Tables 5.2, 5.3, and 5.4 have been updated to explicitly include the relevant test results, interpretations, and discussion. Figures 5.1, 5.2, and 5.3 are added to compare results across conditions.

Hypotheses H1 and H2 are analyzed using a two-way ANOVA framework, and the results are reported and interpreted. I believe these revisions resolve the discrepancy and significantly strengthen the rigor and transparency of the analysis.

7. The title of Section 6.5, “Pilot Testing Digital and Physical Card Game with Nursing Students,” introduces nursing students into the thesis without adequate context or justification. There is no clear explanation as to why nursing students were included in the study or how their participation aligns with the overall research objectives, which primarily focus on CHWs. Moreover, despite the section title, much of the discussion centers on findings related to first- and second-year ANM trainees, with only limited reference to third-year B.Sc. nursing students. This inconsistency between the section title and its substantive content creates confusion. The rationale for

including trainees—whether ANMs or nursing students—has not been clearly articulated, making it difficult to understand their role within the broader research framework.

Response:

Section 6.5: The inclusion of nursing students was exploratory. Nursing students share functional and educational overlaps with CHWs, particularly in frontline healthcare delivery, community engagement, and training needs. The aim was to check the applicability of the intervention across different levels of formal healthcare training. Their inclusion was intended to provide insights into how such game-based interventions could be integrated into early-stage training. Now that the correction has been made, the rationale is sufficiently articulated. The analysis, results, and chart focus specifically on third-year students.

8. In addition to the major concerns outlined above, there are several minor issues that should be addressed before the thesis is considered for the award of the PhD degree. For instance, the name International Institute for Population Sciences should be spelled correctly in the thesis. Acronyms such as ASHAs and CHWs are used interchangeably without clarification, which may lead to confusion unless explicitly defined and distinguished at the outset. I also encountered statements that appear irrelevant to the core research. For example, the sentence “Aparajita Mondal’s contributions to research are not a part of her academic pursuits” does not seem to add any value to the research presented in the thesis. Such statements should either be clarified in terms of their relevance or removed to maintain academic focus and coherence.

Response:

Reference: I have corrected the IIPS full form in reference to NFHS reports.

The use of acronyms like ASHA and CHWs has been made consistent while addressing them.

The committee suggested clarifying the co-authors' contributions to the research. I have moved it from the ethics declaration (Chapter-3) to the Author's contribution section towards the end.